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DEPARTMENT OF BIOENGINEERING

Enhance Virtual Reality Context Aware Robot with Eye-Selection Control

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Submitted in partial fulfilment of the requirements for the MRes degree in Neurotechnology of Imperial College London

CID:01996661

Word Count: 5045

March 2022

Abstract

Robotic control can be leveraged as a substitute for humans with motor disabilities and impairments such as paralysis, paraplegia, limb amputation or ALS to complete a physical task. This human- machine interface which alleviates the dependency on muscle motor control has been combined with various other physiology modalities as an input for robotic control. Previous studies target gaze tracking, pupil dilation, electromyography, electroencephalography, and other biometrics as emerging sensory inputs that seek to interface human with machine. Gaze Tracking has had varied success in robotic control for tasks due to the difficulty to translate this input into low level joint control and the unintentional selection and deselection of point of interest. Additionally, this raises risks of robotic damage when interfacing with costly robotic hardware due to the volatility and inconsistency that may come with biometric signals. To address this complex paradigm, we propose a novel system that targets a simulated virtual reality robot controlled by eye-gaze tracking to complete a pick and place task. This system provides a novel test environment for exploring and improving new methods for gaze selection in a virtual human-computer interface.

Acknowledgements

An immeasurable gratitude is expressed to my supervisor Dr Aldo Faisal for providing me the opportunity to enhance my research skills at the Imperial College London Brain and Behaviour Lab and for being so accommodating to my circumstances amongst an unprecedented time. Additionally, a special thank you to Dr Ali Shafti, for his support and guidance throughout this endeavour and to Chai Auepanwiriyakul who's patience and optimism assisted in producing the final product. Finally, to the Neurotechnology department at ICL for support during this endeavour, my friends and peers at Imperial College, lab colleagues, and family here in the UK who have made this hybrid learning journey all the better.

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List of Abbreviations

AI	Artificial Intelligence
API	Application Programmatic Interface
BCI	Brain Computer Interface
C#	C-Sharp Programming Language
CNN	Convolutional Neural network
DES	Digital Eye Strain
DL	Deep Learning
EEG	Electroencephalography
EMG	Electromyography
ERP	Event Related Potentials
HMD	Head Mount Display
HMM	Hidden Markov Model
HTC	High Tech Computer Corporation
JSON	JavaScript Object Notation
LCD	Liquid Crystal Display
ML	Machine Learning
M	Meters
MS	Milliseconds
2D	Two-Dimensional
Hz	Hertz
POR	Point of Regard
RL	Reinforcement Learning
RNN	Recurrent Neural Network
ROS	Robotic Operating System
RPC	Remote Procedure Call
RViz	ROS Visualizer
SMI	SensoMotoric Instruments
TCP	Transmission Control Protocol
URDF	Unified Robotic Description Format
UR10	Universal Robot 10
UX	User Experience
V-HACD	Volumetric Hierarchical Approximate Convex Decomposition
VR	Virtual Reality

1. Introduction

1.1 Motivation

Assistive robotic computation systems that enable human motor tasks among the physically disabled has long been an area of research that continues to be explored. Injuries to the spinal cord, strokes and neural-degenerative diseases can attribute to the need for such systems. These tasks can be classified into two categories, social interactive robotic systems, or physically assistive systems. (Veneman et al. 2007) Social interactive robotic systems can be exemplified through robots that aid elderly people and specialize in assessing body language, speech, and comprehension. For physical assistive robots, devices can be integrated as cortical limb prosthetics, exoskeletons (Veneman et al. 2007) and rehabilitation interfaces. Gaze tracking is commonly used for human machine interfaces. Recent studies present new methods for 3D gaze point estimation, and its integration with a robotic system architecture. (Shafti et al. 2019) This allows real-time decoding and decision making for the robot to be reactive in respects to the user's gaze.

Gaze selection using eye-tracked gestures and other gaze modalities as a means for robotic control has afforded several challenges. One example of a challenge in gaze-based robotic control that is a widely known limitation is the "Midas Touch" or the "clutch" problem. The Midas Touch problem is a dwell-based interaction resulting from the always-on auxiliary properties of an eye-gaze. The eyes are analog inputs, identifying objects to couple perceptive decisive actions downstream. (Jacob, 1993) This ocular search for inputs can cause unintentional selection if the eye is fixated on a particular point of interest.

This phenomenon impedes the practical use of utilizing eye tracking and gaze selection on robot control. However, gaze coupled with other facial manoeuvres such as blinking, winks, eye gestures or other expression improves the system's ability to differentiate between using the eye input as a cursor versus a selection. Yet this method removes the natural interaction between user and computer and generates complications when newly added gesture actions are needed to select. (Hartson & Hix, 2007) To explore this we seek to provide a system that allows new novel ways gaze selection methods that can be assessed in common robotic control tasks.

1.2 Objectives

The increase in affordable eye trackers and open-source software for gaze interaction has enabled more novel integration of eye tracking into devices and

equipment. More recently eye tracking has become ever more interoperable in robot control. Eye tracking as an input has been commonly used for robot tasks such as locomotion (Seara & Schmidt, 2002) human-robot interaction (Seara & Schmidt, 2003), pick and place of objects and vision control (Kohlbecher et al. 2012)

Movement impaired and healthy people can both benefit from being able to interact with the physical environment (e.g., when operating or interacting with robots) or the virtual environment such as Virtual or Augmented reality worlds. We seek to develop human-robot or human-computer interaction interface that integrates eye movement with multiple methods of "mouse click" for selection to overcome the Midas touch problem. This technology would have immediate impact for both patient and consumer technology. We will develop the project in VR using a virtual robot and physical simulation. To this end we want to build a prototype system of an open-source framework, where we provision a virtual version of our gaze-based assistive robotic system in a targeted game engine for sufficient experimentation on control interfaces and mechanisms that can be interoperable with the robot and enhance gaze-based selection. The initial robotic architecture of the system will be largely based on the Gaze-based, Context-aware Robotic System for assisted reaching and grasping. (Ushida et al. 2006) The delta between the two system will be a robotic gripper that grasps virtual objects on behalf of the user rather than assist their limbs reach and achieve a grasp for the participant using a Carbonhand robotic glove. Our eye-tracking equipped VR system can then be used study and assess performant user control of the robotic system to perform object manipulation tasks in the virtual environment. We will then investigate multiple modalities of gaze selection and observe how performant they are in the completion of various virtual tasks.

As a preliminary iteration of the system to enable a wide range of VR headsets to utilize the framework, we will develop an eye tracking substitute whereby we show a "cross-hair" in the VR system. This is controlled through head movement and is superimposed onto the virtual reality scene. This is an implementation of the "head mouse" (Sim & Faisal, 2006) that will generate initial data identical to an eye tracker. This will allow us to develop all the relevant functionality later overlay the eye tracking data from the API.

1.3 Contributions

Eye tracking as an input modality for virtual reality is a well-established

research space that continues to grow with the increase in cost effective VR gaze enabled HMDs. Additionally, eye tracking as a method for robot control and human robot interaction continues to surface breakthroughs in grasp control of prosthetics (Li & Liu, 2019), exoskeleton navigation and robotic arm control. We seek to merge robotic control through a virtual reality medium to understand how performant gaze selection as a paradigm can be utilized for various tasks.

2. Background

The human visual system under a neuroethological lens, displays a remarkable level of evolutionary ingenuity and coherence. Light is emitted through the retina where the focusing of the light is done in the fovea. Photoreceptors are triggered through rods and cones that fire in the visual cortex of the brain. These receptors assist the brain in cropping light colours into frames and this is done every 13 milliseconds. (Lopez-Caudana et al, 2019)

Evolution has even optimised this system in primate eyes by developing facial attributes to assist with upholding this auxiliary input. Hence the development of various anatomical structures that aid in the protection of the eye and the obstruction of the signal. Eyebrows keep sweat and liquids from rolling into the eye, eyelashes keep dust debris from developing on the sclera and tears remove the excess particles from the surface. A user's gaze is an active input that holds interesting insight into user intent and engagement. (Ademani & Scassellati, 2017)

Our solution can be separated by three components we seek to investigate to achieve our objective: how efficient is gaze selection in VR, what gaze gestures or selection methods can be used for invoking our robot grasping task, and how performant is a simulated virtual robot in executing a pick and place.

Virtual reality headsets contain two LCD monitors that are displayed over each individual eye giving the illusion of depth through a stereoscopic. Improvements in VR seek to reduce the amount of digital eye strain (DES) which is encompassed when the range and depth configured in the head mount display feel incorrectly natural to the user. DES may be and measured using several questionnaires, or evaluations of critical flicker–fusion frequency, blink rate and completeness, accommodative function, and pupil characteristics. (Sheppard & Wolffsohn) 2018

When assessing the performance of gaze selection, we must consider the impact

of eye fatigue in the VR headset, an area of research that continues to be explored to provide more optimal immersive experiences using the device.

2.1 Previous Work

2.1.1 Virtual Reality and Eye Tracking

Repeated investigations in recent literature interrogate and draw into question the effectiveness of eye tracking and gaze selection as a viable human computer interface. Studies on eye tracking as an input alternative continues to be interrogated as some articles find it lesser performant in terms of error rate, throughput, and selection time in VR. In the case of the FOVE VR headset, a study concluded that eye-only input as a method for a selection task was 40% more error prone when compared to head mount only tracking. (Qian & Teather, 2017).

Eye tracking in VR for auxiliary tasks that are similar to a cursor, or analytical feedback prove to be more practical uses for gaze in 2D & 3D immersive experiences. These include "flashlight" to illuminate scenes (Duchowski & Shivashankaraiah, 2000) analysing intent & spatial memory in a scene (Marwecki, &Wilson, 2019), and inspections.

For assessing the success of gaze selection in virtual reality as a viable interface, a new paradigm that can be utilized is Fitts's law. In design, UX & human computer interface, Fitts's law states that the time to acquire a target is a function of the size of the target and the distance to that target.

Figure (1) $$MT = a + b \times \log_2\left(\frac{D}{W} + 1\right) = a + bID$$

Uniformity in leveraging Fitts's law as a metric for assessing eye selection methods in VR has yet to be widely adopted due to extensive variations in tasks and selection modalities that require custom success markers to evaluate performance.

2.1.2 Gaze Selection Methods

Object selection with gaze tracking typically requires some amount of "dwelling" or intrinsic gesture to trigger an action that engages with a target. Dwelling is easy to train as it is reflective of a natural standard delimiter for human gaze to engage with a point of interest. To combat the Midas-Touch problem, developers utilize fixed dwell times to enable users to deselect. This increases the time to completion of a task coarsely diminishing the user

experience due to their unnatural gazes.

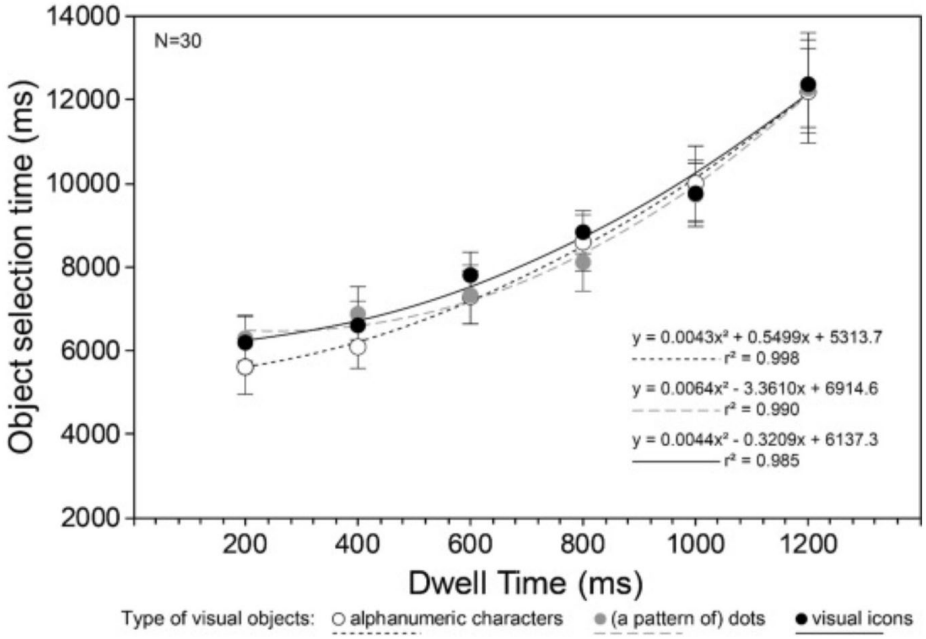


Table 1. A regression analysis of different dwell resulting in a linear selection time of a decoder keyboard speller (Yesaya & Gerard, 2021)

Dwell time typically ranges from 450ms to 1 second but has been improved to around 300ms in some rapid computer interface typers. (Majaronta, 2009) However when the dwell time is too short, users often adjust to the interface by looking away, avoiding accidental selection contributing to the Midas touch.

Dwelling is a fundamental attribute involved with gaze gesture selection as it is in the DualGaze (Mohan, 2018) and SnapClutch (Istance, 2001). These techniques introduce a trajectory adaptive flag that adjacent to the target object to reduce unintentional selection. Though the results of these methods did output reduced selection in a VR task and proved significantly more accurate than fixed gaze (p-0.0004), the task was limited to solely 2D objects in a VR scene. Other gestures when coupled with dwell in 3D gaze selection tasks include duo-reticles, radial pursuit, and nod & roll. (Piumsomboon & Billinghurst, 2017) In this study, head gestures proved the most performant on task selection since long dwelling is unnatural for human eyes due to the normally saccades that occur during gaze.

2.1.3 Robotic Control Selection Tasks

Control of robot movement using a human interface has also been a long well researched area that continues to surface new mediums in robot control. Control can come in the form of gaze selection, non-invasive brain computer interface, human imitation learning, and electromyography (EMG) (Decker, 2008; Stankevich, 2016; Mondini, 2020; Niehorster, 2020). Additionally, simulated robotic control has been achieved in VR for visual odd ball paradigm tasks using P300 from a magnetoencephalogram MEG (Reichert & Rieger, 2014), assistive control with a physical 2D planar robot for a painting VR task (Zhang & Shi, 2021), and gaze & head navigational control of a telepresence robot with an HMD. (Hansen & Zhang, 2018)

The usage of binocular eye tracking which enables distinguishing between naturally occurring involuntary blinks from voluntary winks that do not occur organically has proven to be an effective modality in "mouse click" commands when triggering action to control a robotic actuator and exoskeletons, outperforming selection display units and dwell times. (Noronha & Faisal, 2017) Our research draws heavily on the work of Shafti's context aware assistive robot control through gaze selection (Shafti et al. 2019)

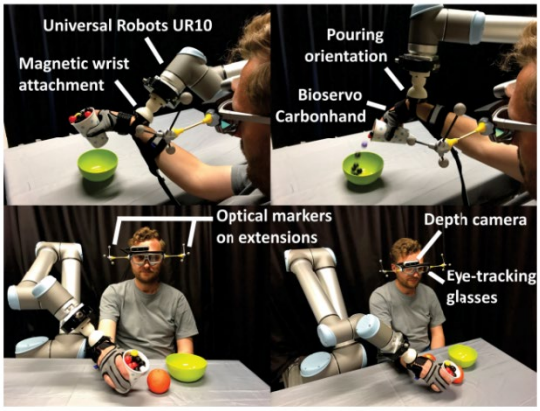


Figure 2. A user with robotic reach and grasp support system. Optical markers in the lens track the subject's gaze and supports the user's wrist in pouring and placing tasks in the table environment. (Shafti et al, 2019)

In this study, example tasks that were conducted involving: pick and place of an object: were run on 5 participants and showed 96% successful implementation of reaching, grasping, and placing on a physical surface based on the 3D gaze of the subject in a physical space. We seek to adopt this methodology to convert the modality to a virtual environment.

2.2 Rationale and Research Proposal

The goal of this study was to investigate the accuracy of gaze selection through a virtual reality interface on a robotic control system. Additionally, we seek to understand what common robotic tasks can be completed in the simulation

framework. This study is an entry point to instantiating an open-source VR environment to explore novel ways for control of robots using gaze selection.

We modelled our VR environment based on the Universal Robot 10 pick and place task experiment studied in Shafti et al. The UR10 leverages the Robotic Operating System, a widely adopted software library for building robotic platform applications and vehicle control whose framework has been extended to connecting the GAZEBO simulator with the Unity engine simulator for virtual functionality. (Hussein & Olaverri-Monreal, 2018)

Our objective was to identify the optimal dwell time that abides by Fitts's law for human interface design that achieves a seamless selection of an intended target for the virtual UR10 robot actuator to grasp and place in the scene. Moreover, no research uses gaze selection in a fully virtual reality environment for control of a virtual robot.

Our study seeks to evaluate to success and performance of dwell time gaze selection in VR for different target distances and its interoperability with a simulated robotic control system to grasp, pick and place a targeted object in a virtual scene.

$$H_0 = \mu(Longer\ Dwell\ Times)errors < \mu(Shorter\ Dwell\ Time)errors$$

3. Methodology

3.1 Head-Mount Display Specification

This study used a 2016 Tethered VR HMD based on the HTC Vive with an integrated binocular eye tracker from SensoMotoric Instruments (SMI). The device comes packaged with a Unity Engine API written in C# for accessing streaming data from the SMI server. Unity is a cross-platform game engine provided by Unity technologies that provides a platform for constructing 3D/2D games. The SMI eye-tracking outputs data points at a frequency of 250Hz but is collected and captured in Unity at a frame rate of 100Hz which is mapped to the complete resolution of the HMD screen (2160x1200). The origin is located at the top left corner of the display and contains a field of view of 105.6 ° vertical and 111.9 ° horizontal.



Figure 1. Tethered Eye Tracking Virtual Reality Head Mount Display based on HTC Vive Cited from (Gaze Intelligence 2016)

Although it is no longer available for sale, the SMI Eye Tracker was selected as our Eye Tracking HMD in VR due to the Brain and Behaviour Lab's previous utilization of the device in recent studies and the HMD's ability to outperforms other HMDs available on the market at the time of writing. Data quality measures from (Niehorster et al, 2020) plot the gaze recording and device slippage that identify the SMI with the highest Accuracy (Deviation) with the lowest data loss.

3.2 Simulation Robot Configuration

The experiment targeted the Universal Robot 10 (UR10) as the simulated robot arm to be controlled in the study with an attached Robotiq 2F-140 gripper as the end-effector link, modelled after the physical robot used in the context aware assisted reaching robotic system from (Shafti et al. 2018). The simulation robot was controlled using ROS (Robot Operating System), an open-source library framework for kinematic modelling, pose path planning and message remote procedural calling (RPC). We used the Felix Exner maintained repository of the ROS-industrial universal robot meta-package.

To optimize stability and enable mobility between local and lab machines, this configuration of ROS was packaged into a docker image. Unity Engine is primarily stable on Windows operating system and ROS is well documented and supported on Linux. Offloading the ROS trajectory movement planning for the robot kinematics was isolated in a docker container to minimize compatibility issues between Unity Engine and the ROS environment with their respective dependencies. The generated docker image used the ROS melodic morenia distribution with custom service files for synchronous blocking and request response communication between Unity and ROS.

The study used catkin tools version 0.6.1, a low level build system macros for building ROS projects.

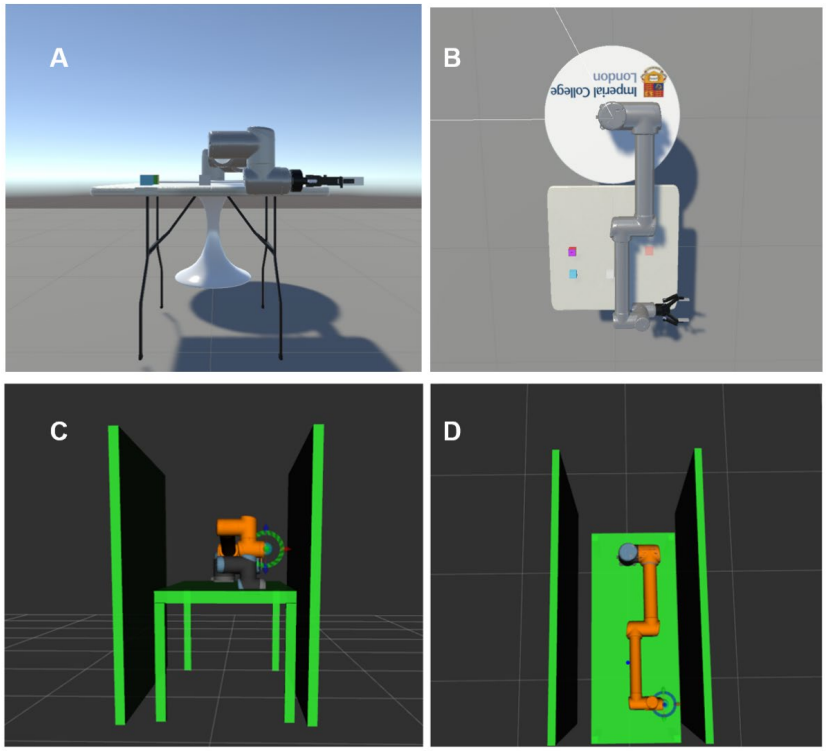


Figure 2. Simulated UR10 with Robotiq 2F-140 A) Subject Camera view of rendered UR10 Robot in Unity Scene (B) Aerial of rendered UR10 Robot in Unity Scene mounted on a table at neutral

state. (C) UR10 Robot frontal view in RViz simulation with scene objects used to constrain the motion planning of the robot. (D) UR10 aerial view in RViz simulation with scene table and walls.

ROS Scene objects were added through the Planning Scene ROS API to assist with trajectory planning. A table of 0.7x1.7x0.7(meters) was added as an attached object to the UR10 on the mover service node. To prevent the robot from achieving obtuse trajectory poses, two ROS scene walls were added to constrain the UR10 to a smaller movement area.

3.3 Unity Robotics Hub & The Articulations Engine

Unity Robotics Hub is a recently released central library repository we utilized to simulate the UR10 in the game engine. It requires Unity version 2020.3.11f1 which we target in our project. The hub uses the PhysX 4 simulation SDK by Nvidia that introduces rigid body dynamics in the game engine. Unity's solver for small physics engine tasks that calculates joint interactions within the robot for determining collisions uses the Temporal Gauss Seidel or Projected Gauss-Seidel-style calculation. Our experiment project uses the ROS TCP Endpoint Connector v0.5.0, the URDF Importer v0.5.5, and the SteamVR Unity Plugin version 2.7.3 for HTC Vive control.

The ROS TCP Endpoint package bridges the communication between our Unity Scene and the ROS motion planning for the simulated robot. The connector wraps the message for published positions in the Unity Engine scene as a TCP client for ROS to calculation a path plan for a pose trajectory. Trajectory execution is handled in Unity by the Articulations Engine. Figure 3. Outlines the serialization of messages between the ROS Network and Unity.

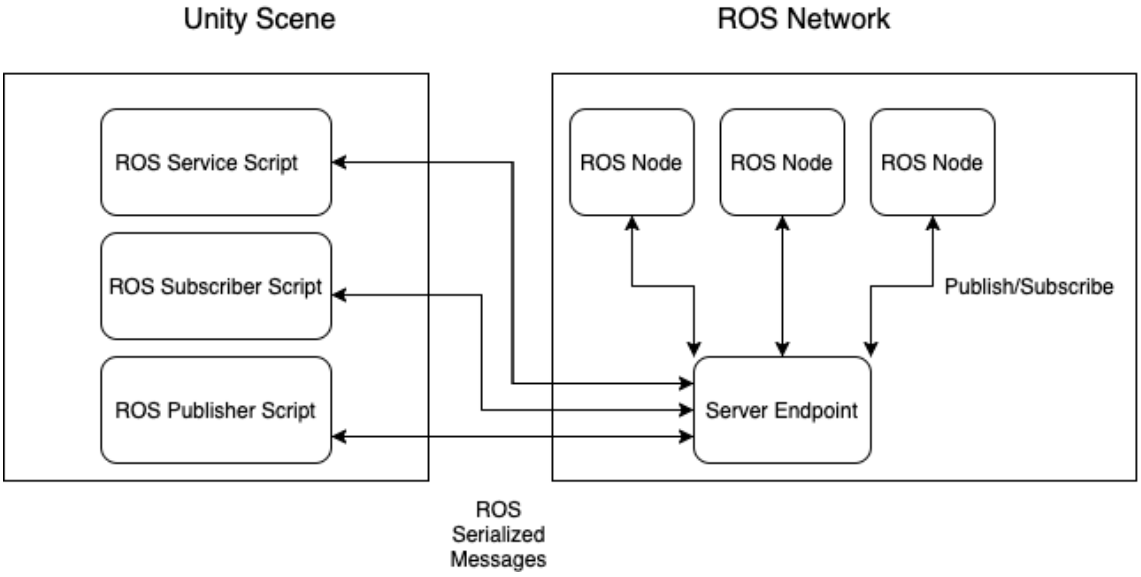


Figure 3. A ROS node running the TCP endpoint, facilitates bidirectional messages to and from Unity and ROS. The Messages are serialized using the Message Generation plugin that creates C# scripts with the ROS .msg files for serialization and deserialization. Retry logic is embedded in the connector for lost connectivity and passing the ROS prefab instance in between Unity scenes. Adapted from (Unity Technologies. Unity Robotics Hub, 2021)

The UR10 and Robotiq 2F-140 gripper use a custom URDF we configured that has disabled additional collision links within the gripper and the robot joints to decrease the amount of trajectory execution artifacts using the MoveIt! Setup Assistant. For increased accuracy with collision detection, we use the URDF Importer with V-HACD mesh decomposition for more realistic interactions in the Unity Engine scene.

Once imported into the Unity scene, the base link of the UR10 robotic arm was mounted to the robotic arm table as an immovable object. The stiffness of the joints was set 10,000 units and the damping of the URDF was set to 1000 units to avoid robot link slippage. Additionally, the force limit was set to 1000 units along with the speed and torque set to 30 units and 100 units respectively to avoid a high velocity change in applied force to achieve a desired pose position. The articulation body allows the robot joints to be controlled by positional, torque or velocity control. The articulation body controls all utilize the Spring-Damper equation to determine the force applied on the robot joint. The equation in respects to the robotic arm is provided below where p = current position, p' = target position, v = current velocity and v' = target velocity

$$Force = stiffness \cdot (p - p') \cdot damping \cdot (v - v')$$

3.4 Experiment Design

The experimental design is shown in Figure 4, Figure 5, and Figure 6. The Unity scene was modelled akin to the fully integrated robotic system test setup from Shafti et al. The difference being the UR10 Robotic Arm is mounted across from the subjects view in the simulation scene for independent pick and place as opposed to assisted pick and place tasks.

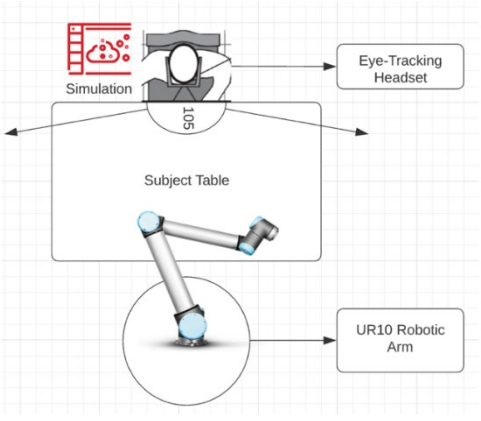


Figure 4. Diagram displaying the participant location to the UR10 Robotic Arm in the simulation with table scene objects. (Not drawn to scale).

Pick and place is a common robotic arm assessment task to validate the integrity of a robot interface system. Shafti et al (2019), explores hierarchal action semantics around pick and place on the table, pick and place into containers and pick and pour into larger containers. For simplicity of our simulation, we will seek to have the participant invoke the robot to invoke a pick and place on predetermined target goal placement. To assess our system, our performance markers consist of the participants correctly invoking a pick and place on a targeted object, the duration of time for the user to select an object, and finally, the simulation robot's ability to achieve a pick and place of the intended object in proximity to the goal placement area.

Three 5x5x5(cm) cube blocks weighing 1-gram units in the game engine were generated on a virtual table in front of the participant. The size of the blocks was selected to target fitting two blocks comfortably in the Robotiq 2-Finger 140 gripper opening of 14cm. (Robotic Inc, 2018) Additionally the block dimensions displayed identical properties to the apples and oranges used in the context aware robotic system. One block was designated as the 'target block' to the subject with multi-coloured surface characteristics (red, blue, yellow). The other two blocks were decoys with neutral dark yellow colours to give the participant added obstruction to the target block and increase difficulty of selection surface area. All blocks were given the property of yellow feedback once the participant's eye gaze fixated on any of the block to provide visual context to their gaze location cursor. Finally, a partially opaque red rectangular prism was used of size 8x6x2(cm)as the target placement object. The rigid body component was removed to prevent collision with block objects as we observe the distance between out block objects and goal to determine UR10 pick and place success.

The subject table in the unity scene renders at 80x80x60(cm) with a square 2D 80x80(cm) surface the blocks are generated on. The entire table remained contained within the UR10's reachable grasping area at the robot table's starting position. The total experiment placement area in which all blocks and target placement spawn at the start of each trial was measured at 70x70 cm on the table surface area. An added 10 cm was given between the experiment working area and the table surface edges to prevent blocks from respawning during a trial start and falling off the edge of the virtual table.

We developed a target quadrant system of 4 smaller squares on the virtual table

to distribute the placement of the target block to examine the effect distance from the user has on selection time and distribute a relative target position placement among all participants. Each respective quadrant was 35x35cm and the target block position was randomly spawned in a selected target per trial. Figure 5 shows the quadrant system and experiment area on a grid in respects to the surface table.

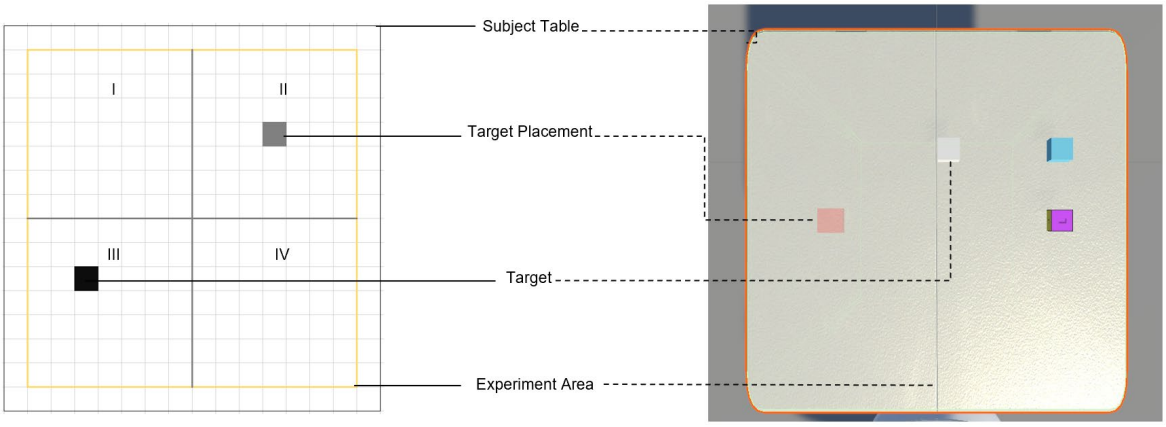


Figure 5. The left grid displays the distribution of the Experiment Area with embedded quadrants on a grid. The right image is the rendered surface table from an aerial view with the three blocks and the target placement area.

The Enhance VR Pick and Place experiment consisted of 4 task blocks in which a gaze selection modality by the user was observed over 12 trials. The four gaze selection blocks included dwell times of: 0.5, 1.0, 1.5 seconds and a manual 'mouse click' selection type as our control. For our success markers, we observe the selected target with task gaze action to calculate the time to invoke the given task. The participant was told at the start of the task block how to select that target cube based on the gaze selection type with the dwell time omitted. The sequence of the task block as they were completed in the experiment was randomized per each participant.

Enhance VR Context Aware Robot Pick and Place with Gaze Selection							
Task Block	Dwell 500ms		Dwell 1,000ms		Dwell 1,500ms		Mouse Click
Action	Select Target with Task Gaze		UR10 Pick Target			UR10 Place Target	
Trial States	Idle	Task Block Selection	Idle	Trajectory Plan	Pick up	Place	Output Data

Figure 6. A Topographical outline of the experiment procedure with four task blocks where the Gaze selection type is inputted as the participants goal for each trial. The actions define the three activities during a trial where the subject must complete the first action of selecting a target and the robotic arm completes the second and third action of pick and place. The trial states are the lower-level logic that determines what input is required to progress the state to the next action.

After 12 trials, the task block concluded, and the participant was asked three questions to assess user's experience and perceived workload with accomplishing the task block. We used a 5-point scale based on the NASA Task Load Index (Hart S.G. 2021). The questions targeted the mental demand, perceived success, and performance effort to complete each individual block segment in the experiment

We asked the participant to stay seated during the experiment with the option of manoeuvring within their seat to complete their task. A physical table was placed in front of them relative to the virtual table at 35 cm from the centre of the chair. The table was used to place the computer mouse on a surface for a fluent experience in selection during that task block.

Prior to the start of the experiment, user calibration was done for each user individually on the SMI Eye Tracking HTC Vive HMD. We conducted a 5-point calibration which required the user gaze to follow the motion of a point displayed in the head mount display. Five-point has the highest gaze accuracy on the device with calibration targeting up to 1.5 m in distance within the Unity scene (Gaze Intelligence, 2016) Calibration was invoked by toggling the (4) key on the lab machine keyboard after starting the scene and the SMI server was successfully initiated. Each time the participant placed the HMD back on, they were asked to align their eye reticule to a red graphic at the centre of their view to adjust the HMD correctly on their head.

To provide the most immersive experience, the VR prefab camera was configured at 1.5 meters from the UR10 base in the unity scene. The HMD screen field of view was set to the default 60° and enabled rotational & positional tracking type with the Vive. This allowed the user to rotate their head in the VR environments and lean to improve their angle at achieving a gaze selection.

After observing the participants completion of the task segments, we noticed a wide range of variability in ways a user sought to achieve fixating on the target goal. Figure 7 shows the point of view from the observers' machine that receives the HMD feed of the participant. In the sequence of images, we can observe the selection of the target block, the pick, grasp, and place of the UR10 and finally the respawning of all the blocks on the virtual table surface.

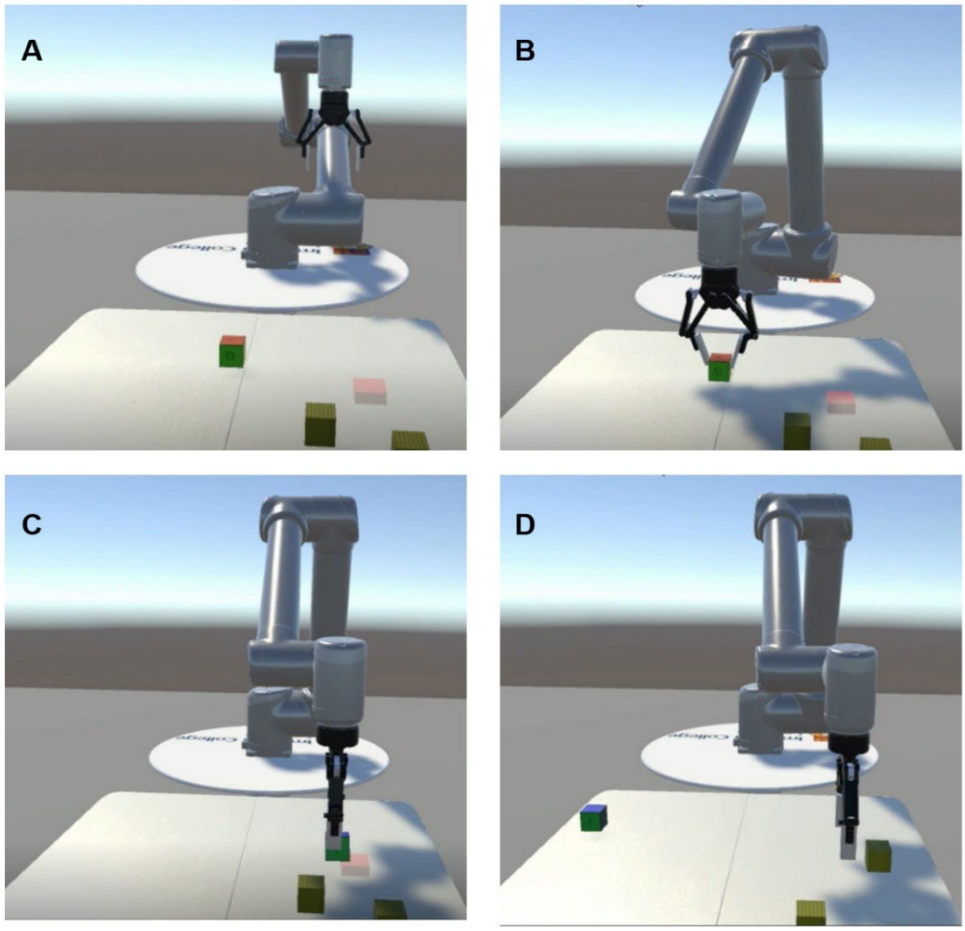


Figure 7. Observers view of the participants ocular stream from the HTC Vive. A) Displays a trial frame at the time of a successful gaze selection. Here the multi-coloured target block in Quadrant I has a yellow hue to signify the current gaze from the subject is fixated on it. B) Displays a trial frame at the pickup trial state for the UR10. C) Displays a trial frame right before the completion of the place trial state for the UR10. D) Displays a trial frame of a newly generated block trial where the target has respawned back to Quadrant I and the decoy blocks have respawned as well. The target now has a normal hue signifying the subject is no longer gazing at the block.

3.5 Data Collection

Trial data was written to a JSON file per participant after each trial. Our code overloads two C# class objects to later be JSON serialized downstream and appended to the current trial JSON file. The SMI gaze data as mentioned prior, outputs eye tracking data at a frequency of 250Hz but is captured in Unity at a frame rate of 100Hz. We process multiple gaze data types for overlaying on the Unity Engine scene. We obtain the camera Raycast, the gaze direction, gaze base point direction (a unit vector of the actual direction the eye is looking in the physical world – not to be used in the virtual world) and finally the left and right eye points of regard (POR).

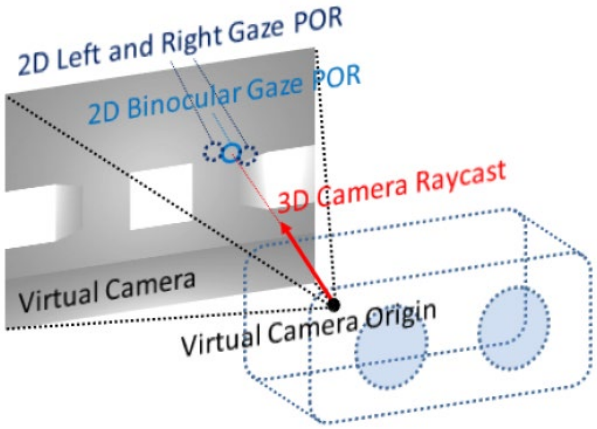


Figure 8. Diagram of the Right Gaze Point of Regard, a 2D vector that is mapped to the complete resolution of the HMD screen, and the 3D Camera Raycast, the aggregation of both left eye and right eye POR overlayed into the virtual camera 3D world. (Gaze intelligence, 2016)

Two series of experiments were completed with the first series of experiment results being dismissed from our final outcomes due to inconsistencies in methods used to achieve that first series output. Additionally, the first series of experiment data lacked the SMI gaze data which tracked a subject's eye fixation during a trial.

The second experiment series included seven volunteers from Imperial College London (7 males, ages m=28.71, S.D.= 4.05). Five participants wore prescription glasses which were removed when the HMD was worn. All the participants were from Imperial College London Brain Behaviour Lab and were chosen based on convenience.

Our data frame used to analyse our results consisted of 40 datatypes detailed in Table 2. Due to the vastness of data points, many views into our experiment data could have been generated but due to time constraints, we resorted to the critical materialized views that captured our success mark of the system.

Type	*Name*	*Details*
Timestamp	Experiment Time	Time at start of Experiment Block
Int	Trial Number	Trial number in Task Block
String	Task Type	Gaze Task for Pick and Place
String	Subject Name	Participant Name
Int	Subject Age	Participant Age
Float	Subject Height (cm)	Participant Height
String	Selection Type	Gaze Selection Type
Float	Selection Time (s)	Time to invoke the selection of table object
Bool	Selection Success	Boolean if Target block was selected
Float	Total Execution Time (s)	Time to complete Pick and Place
Float	Articulation Execution Time (s)	Time UR10 completes trajectory execution
Float	ROS Execution Time (s)	Time for ROS to publish a cartesian plan
Vector3	Target Position Initial	Target Block Position at trial start
Vector3	Target Position End	Target Block Position at trial end
Float	Target Gaze Time (s)	Total time Subject gazes at Target
Float	Target to Placement Initial Distance (m)	Target Distance to Goal at trial start
Float	Target to Placement End Distance (m)	Target Distance to Goal at trial end
Float	Target to Placement Delta Distance (m)	Delta of Target Distance to Goal during trial
Int	Target Quadrant Index	Target Quadrant indexed location
Vector3	Block I Position Initial	Block I Position at trial start
Vector3	Block I Position End	Block I Position at trial end
Float	Block I Gaze Time (s)	Total time Subject gazes at Block I
Float	Block I to Placement Initial Distance (m)	Block I Distance to Goal at trial start
Float	Block I to Placement End Distance (m)	Block I Distance to Goal at trial end
Float	Block I to Placement Delta Distance (m)	Delta of Block I Distance to Goal during trial
Vector3	Block II Position Initial	Block II Position at trial start
Vector3	Block II Position End	Block II Position at trial end
Float	Block II Gaze Time (s)	Total time Subject gazes at Block II
Float	Block II to Placement Initial Distance (m)	Block II Distance to Goal at trial start
Float	Block II to Placement End Distance (m)	Block II Distance to Goal at trial end
Vector3	Block II to Placement Delta Distance (m)	Delta of Block II Distance to Goal during trial
JSON Array	Gaze Points	Array of SMI Eye Tracking fields to parse
Timestamp	Gaze Points – Gaze Time	Unix time stamp of SMI gaze points collected
Vector3	Gaze Points – Gaze Position	Raycast vector position of SMI gaze

Vector3	Gaze Points – Gaze Base Position	SMI Gaze base point vector at left eye centre
Vector2	Gaze Points – Gaze Left POR	SMI left eye point of regard in 2D frame
Vector2	Gaze Points – Gaze Right POR	SMI right eye point of regard in 2D frame
Float	Gaze Points – Gaze Frequency	Static frequency of published SMI data (100hz)
Bool	Selection Failure	Conditional if Selection time is <= 0 or Selected incorrect block object
Bool	HMD Signal Loss	Dropped connectivity HTC Vive base w/ HMD

Table 2. Enhance VR Context Aware Robot Dataset Descriptive Chart

For our success metric, we decouple the components of our trial experiment between the Gaze Selection and the Robotic Arm execution described in Table 3. For user's gaze ability to trigger the robot, we use the Selection Time, Selection Success and Selection Type (i.e., Dwell times). For assessing the robotic arm execution and the integrity of our ROS environment interfacing with unity, our values for ROS Execution Time, Articulation Execution Time and Target Placement End Distance determine our rate of success.

Component	**Data Fields**	**Description**
Gaze Selection Assessment	Selection Time	Time it takes to gaze select the intended block
	Selection Success	Bool to check if Target Block was selected
Robotic Arm Assessment	ROS Execution Time	Time for ROS to generate a trajectory plan
	Articulation Execution Time	Time for the Articulations Engine to execute plan
	Target Placement End Distance	The target block distance at end of trial from target placement

Table 3. Enhance VR Context Aware Robot Success Metric Descriptive Chart

3.6 Data Analysis

Data was aggregated per each subject using a IPython Jupyter notebook to append the data frames per the respective task blocks. ANOVA analysis with repeated measures of experiment outcomes and data cleansing were completed using Microsoft Excel, PANDAS and scikit learn. Power BI was utilized for graphical chart views of the data.

4. Results

4.1 Participant Assessment of Selection Tasks

After the conclusion of each task segment block in a participant experiment, the subject completed a 3-question assessment regarding the difficulty aptitude and mental demand scoring of the previous task. Table 5 displays a Likert Scale of 8 subject responses which depicts the dwell task of 1.5 seconds having the least rated user experience regarding mental demand, participant success rate sentiment and participant effort tin the completion of the task.

	Manual Mouse Click			Dwell 0.5(s)			Dwell 1.0(s)			Dwell 1.5(s)		
	Frequencies	Median	SD	Frequencies	Median	SD	Frequencies	Median	SD	Frequencies	Median	SD
How mentally demanding was that task?		4.5	1		5	1		4.5	1		3.5	1
How successful were you in accomplishing what you were asked to do?		4	1		4	1		4	1		3	1
How hard did you have to work to accomplish your level of performance?		4	1		4	1		4.5	1		3	1

Likert Scale
1 2 3 4 5

Table 5. Displays the density function visualization that plots Subject task experience on a 5-point Likert Scale based on NASA Task Load Index for each task block. Included is the median and standard deviation. Subject experience decreases for sample during dwell time of 1.5 seconds.

4.2 Gaze Selection Performance

Since our initial study had a small sample size which drastically reduces the statistical aptitude of the experiment, our selection performance methods are not favourable for our null hypothesis. Additionally, data loss that occurred in the retrieval of JSON and SMI gaze points contributed to loss of trials for subjects, this is further discussed in the Limitations and Known Issues section. Nevertheless, we were able to examine interesting feedback results from the 330 trials across 7 subjects conducted.

In Figure 9, an analysis of the selection time was conducted to determine the performance of the selected gaze modalities in our VR simulation. Our distribution of selection time in (a) shows a right skewed log normal multimodal distribution for our trials. When plotted through logarithmic scaling in (b) the shape is maintained.

Of the 330 trials, 14 (4.24%) were incorrect in selecting an alternative block in the experiment by a subject. In (d), the count of failed selection includes the 14 incorrect selections by the user as well as instances the selection mechanism failed with a selection time of 0.00 seconds (16 failures).

Quadrants II & IV were equal in t=rank for the most failed selection trials and the gaze selection modality of 0.5secs contained the most failed trials with 13

(81.25%) of all failures in our sample. The gaze selection modality with a dwell time of 1.5 seconds contained 0 failed trials within the sample.

The average selection time respective to the gaze selection modality shows manual trigger coupled with gaze and a dwell time of 0.5 seconds with the lowest average collectively. Moreover, the target quadrants with the smallest average for selection time were quadrants II and III. This conflicts with our null hypothesis as the distance between the HMD and the respective targets were highest in quadrants I & II.

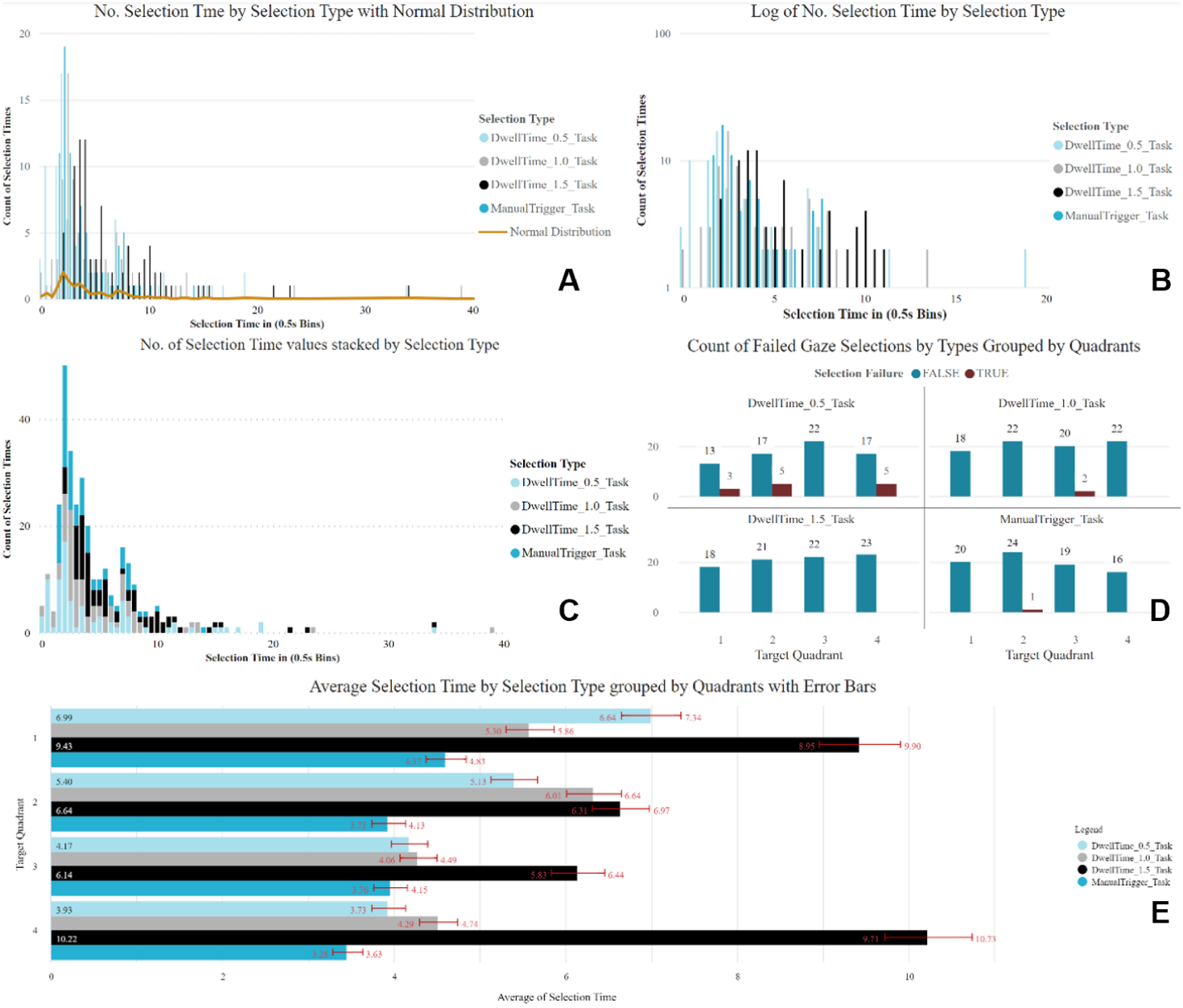


Figure 9. Includes different views of experiment selection time histograms with normal distribution (a), logarithmic scaling (b), and a stacked materialized view of selection time values grouped by the gaze selection

type. Additionally, the selection failures across the four quadrants are plotted in a clustered column chart with small multiple grids of the selection type (d) and the average selection time by selection type in respects to the quadrants with included error bars (e).

4.3 UR10 Robotic System Performance

The system performance varied in terms of component success and execution of the virtual UR10. ROS maintained a relative stable and consistent performance with an average of 2.49 seconds to complete a cartesian trajectory plan for the UR10 to achieve a grasp on the target from the docker image. While the median execution time to grasp and place for the robot was 6.23seconds.

The distance from the selected object to the targeted placement location was quite unstable which at the conclusion of our experiment we determined there are not enough parameters to confidently assess the performance of the UR10. During the execution of the pick and place task, dropping, collision and obtuse poses occurred causing varied results in the robot performance. In figure 10(b) the are between the init and end lines, display the change in position achieved by the robot. In several trials no change was made in this subset sample.

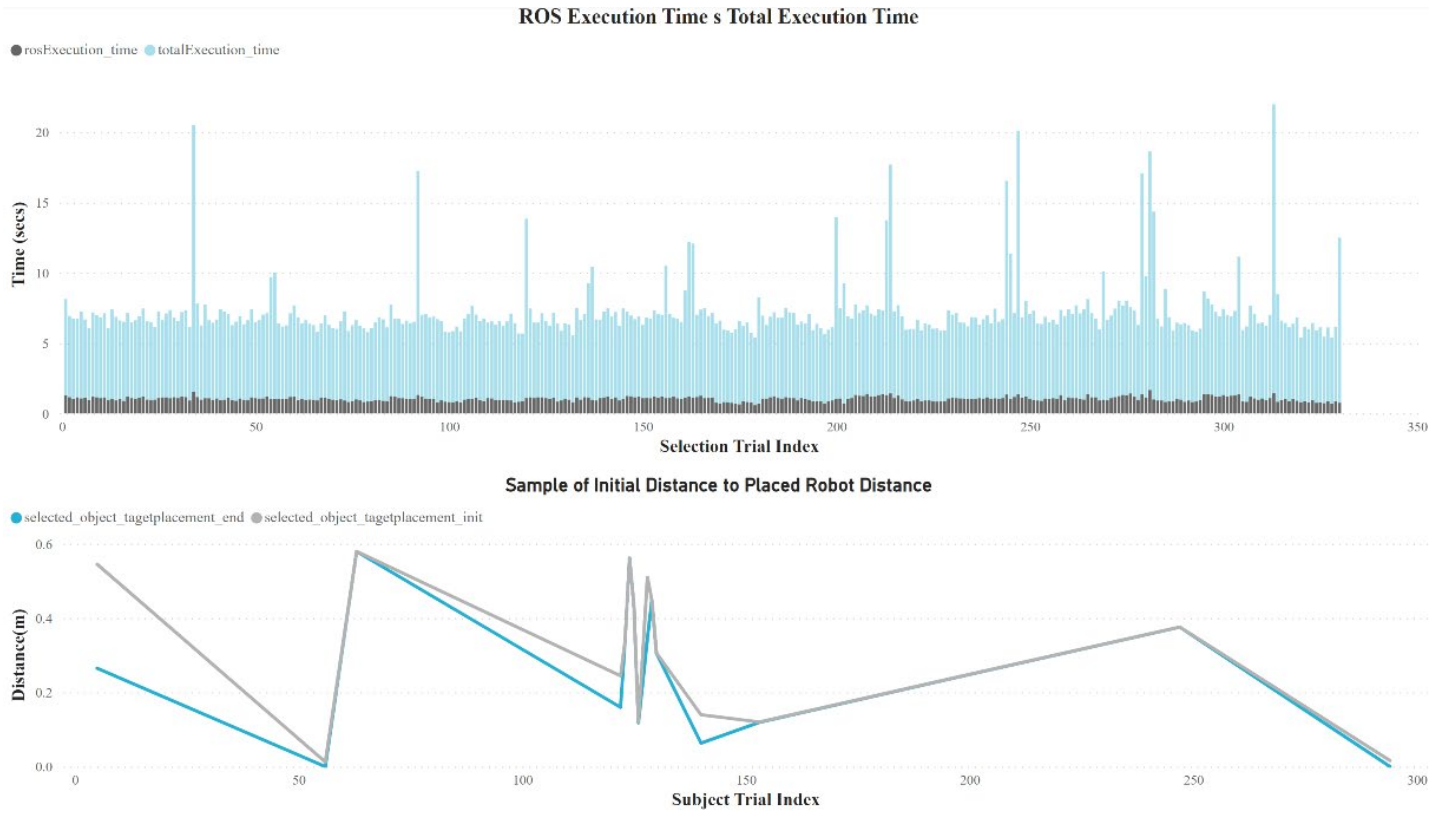


Figure 10. Displayed in (a) is the time duration for the Robotic Operating System (ROS) to complete a trajectory plan vs the duration for the virtual UR 10 to complete the pick and place task. Plotted in (b) is a sample of 16 trials that displays the initial distance of the selected target to the target placement goal.

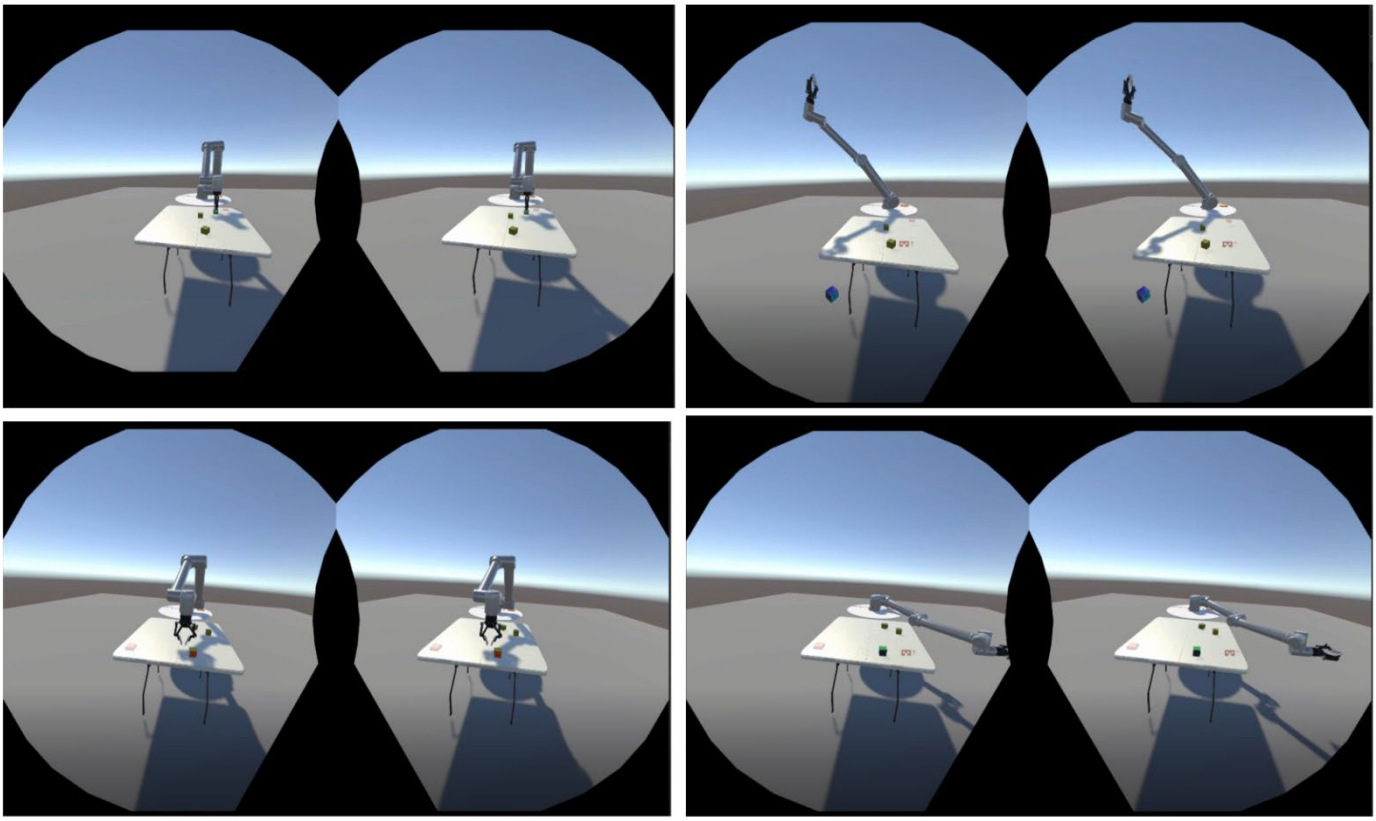
4.4 Limitations and Known Issues

selection_type	selection_success	selection_time	selected_object	totalExecution_time	articulationsExecution_time	rosExecution_time
DwellTime_1.0_Task	TRUE	1.702191	Target	5.78866529	4.85667324	0.9319922
DwellTime_1.0_Task	TRUE	2.54855919	Target	5.278981	4.36432362	0.9146578
DwellTime_1.0_Task	TRUE	2.94293213	Target	8.048006	7.21889639	0.8291097
DwellTime_1.0_Task	TRUE	4.103922	Target	5.96859741	5.08100367	0.8875938
DwellTime_1.0_Task	TRUE	2.604502	Target	5.024606	4.16502142	0.859585
DwellTime_1.0_Task	TRUE	2.46923757	Target	5.438843	4.421788	1.01705456
DwellTime_1.0_Task	TRUE	3.42392254	Target	5.361048	4.37589836	0.98514986
DwellTime_1.0_Task	TRUE	3.40103531	Target	5.604367	4.77212954	0.832237244
DwellTime_1.0_Task	TRUE	6.30282736	Target	5.365732	4.45133448	0.914397836
DwellTime_1.0_Task	TRUE	8.219449	Target	5.085933	4.26398039	0.821952641
DwellTime_1.0_Task	TRUE	A 0	Target	5.4308424	4.501841	0.9290014
DwellTime_1.0_Task	TRUE	0	Target	5.083438	4.13016939	0.953268647
DwellTime_1.0_Task	TRUE	2.937614	Target	4.959288	4.115178	0.844110131
DwellTime_1.5_Task	TRUE	9.901656	Target	5.088951	4.244862	0.844089
DwellTime_1.5_Task	TRUE	10.09283	Target	5.360754	4.526249	0.83450526
DwellTime_1.5_Task	TRUE	2.00195313	Target	7.286517	5.893177	1.39334
DwellTime_1.5_Task	TRUE	4.120818	Target	6.77773666	5.390368	1.38736892
DwellTime_1.5_Task	TRUE	4.19946575	Target	6.42427731	5.085061	1.33921647
DwellTime_1.5_Task	TRUE	3.58197856	Target	6.029229	4.82366562	1.20556343
DwellTime_1.5_Task	TRUE	3.576491	Target	5.749294	4.567834	1.18146
DwellTime_1.5_Task	TRUE	3.440938	Target	6.137077	4.82655859	1.31051838
DwellTime_1.5_Task	TRUE	3.66530013	Target	5.807626	4.59920549	1.20842016
DwellTime_1.5_Task	TRUE	3.279855	Target	5.6431036	4.394393	1.24871063
DwellTime_1.5_Task	TRUE	4.332905	Target	5.99170351	4.694855	1.29684842
DwellTime_1.5_Task	TRUE	5.976036	Target	9.809628	8.461312	1.348315
DwellTime_1.5_Task	TRUE	6.2170496	Target	6.62319	5.24960566	1.37358415
DwellTime_1.5_Task	TRUE	14.1529083	Target	B 14.5298243	C 13.0714512	1.45837319

Table 4. Displays example of 25 rows from our experiment output dataset uncleansed partitioned for subject 2. A. displays an instance where the selection time is 0 and the selection success if true exemplifying an instance of a data failure artifact. B. Displays a ROS total execution time of 14.52 seconds, significantly higher than other trials. C. Signifies an instance of a long duration of the Articulations Engine in unity executing the ROS trajectory plan. This trial resulted in a failed pick and place task.

Various failed errors were accumulated in the system causing an investigation of the integrity and reliability of the Enhance Context Aware robotic system to be difficult. ROS communication between the Unity TCP Endpoint and a inaccurate execution of the articulation caused the simulated robot to create obtuse and colliding poses to achieve a pick or place pose.

When calculating the published target pose for the UR10 to achieve for pick and place, we utilize the MoveIt! cartesian plan method which is leveraged to avoid obstacle collision when manoeuvring the robotic arm. This coupled with our instantiated scene objects was our attempt to adequately avoid object collision in the unity scene. However, the executed plan still managed to collide with



objects, nevertheless.

Figure 11. Top Left displays the UR10 prior to executing a ROS MoveIt! plan in a trial. Top right displays the UR10 achieving a pose during its plan execution dropping the target block and eventually colliding with the subject table. Bottom left displays the UR10 prior to executing a ROS MoveIt! plan in another trial. Bottom right displays the UR10 colliding with the subject table at the upper arm link never achieving a successful pick pose.

4.5 Discussion and Future Work

A large proportion of this project completion was dedicated to system development and stabilization of the framework to achieve the reported outcome. A thorough examination of the integrity of ROS-Unity integration and our gaze selection is still in need of completion due to unforeseen bugs that impeded on an accurate assessment of the system performance. For future work we seek to implement data point improvement to further understand what additional VR environment artifacts impact the selection through gaze and ROS

control for the completion of the tasks.

These improvements may include:

- detection of UR10 obstruction between user gaze and the intended targeted block

- removal of Vive idleness which shuts down the HMD when a participant is idle for too long (or the base stations is unable to detect the HMD)

- improved trajectory planning to constrain robot maneuvers from colliding with objects

- the removal of trial logic bugs such as 0 selection time gaze triggering which accounted for 3 trials in our dataset

Another enhancement to the framework we can pursue is pose estimation for the grasping by the Robotiq gripper to optimally pick up the target block using tactile servoing. On multiple trials throughout the experiment, the execution metric of our system failed because the UR10 inadequately grasped the block in the VR environment to apply enough force to carry the target block to the intended goal. Utilizing available training datasets for RGB images, we can train a model through deep learning to predict the pose of the cube and grasp it adequately.

Further development can introduce new tasks such as free form pick, point and place (participant determines the goal placement of the block), pouring, navigation and novel gaze selection modalities. Ideally, we would seek to introduce EEG as a biomarker to investigate the P300 signal utilization for determining intent and selection once coupled with gaze fixation as used in Reichert & Rieger (2014).

5. Conclusions

The human visual system under a neuroethological lens, displays a remarkable level of evolutionary ingenuity and coherence. Our solution to provide a novel way to assess robot simulations using gaze selection had varying degrees of success, but predominately our system proved unstable.

Due to the expansiveness of our original endeavour, our building of our system in Virtual Reality with interoperability between the SMI HTC Vive Eye Tracking HMD and the ROS Universal Robot System our conclusive assessment of the system was constraint to limited success markers. The emerging technology that the ROS-Unity framework has provided as an interface produced substantial instability between publishing a trajectory plan from ROS and an execution plan within the Articulation engine.

Additionally, the 2016 HTC Vive base camera also produced artifacts that muddied the data in which instances of HMD connectivity loss was captured causing an increase in selection time for several subjects.

Finally an identified bug within our algorithm that runs the experiment introduced an instance in which subjects would produce a "phantom selection" in which the UR10 Robotic arm would return to the previous trial after respawning a new trial due to the subject fixating on the position of the cube during the pick and place state. This was introduced during a refactoring of the code and caused a disentanglement with our data that produced a biased result. Since this instance, we have rerun experiments and seek to provide the results of a cleaner dataset within the future. This is an initial step in providing a new cost effective way to circumvent the use of physical robots to explore the Midas touch gaze selection paradigm that limits eye tracking as a reliable input for "mouse click"

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Appendices

Appendix A: Ethical and Legal Considerations

The Enhance VR Context Aware Robot with Eye Selection Experiment involved 8 participants who completed the four gaze selection tasks in VR. The participants above the age of 18. All participants were enrolled students at Imperial College London and were chosen based on convenience. They were asked to participate in the experiment verbally and a signed consent form war obtained on all of the participants.

Observations from these experiments were recorded then converted into anonymous data points. The data has been obstructed to prevent subject identification. Videos were taken of all subjects participating in the experiment with the intent to embed a capture of the tasks completed for later presentations

Besides the aforementioned, this project does not touch upon any of the other clauses on the ethics checklist.

Appendix B: Ethics Checklist

Ethics Checklist Sections	Yes	No
Section 1: HUMAN EMBRYOS/FOETUSES		
Does your project involve Human Embryonic Stem Cells?		X
Does your project involve the use of human embryos?		X
Does your project involve the use of human foetal tissues / cells?		X
Section 2: HUMANS		
Does your project involve human participants?	X	
Section 3: HUMAN CELLS / TISSUES		
Does your project involve human cells or tissues?		X
Section 4: PROTECTION OF PERSONAL DATA		
Does your project involve personal data collection and/or processing?		X
Does it involve the collection and/or processing of sensitive personaldata (e.g. health, sexual lifestyle, ethnicity, political opinion, religious or philosophical conviction)?		X
Does it involve processing of genetic information?		X
Does it involve tracking or observation of participants? It should be noted that this issue is not limited to surveillance or localization data. It also applies to Wan data such as IP address, MACs, cookies etc.	X	
Does your project involve further processing of previously collected per- sonal data (secondary use)? For example Does your project involve merg-ing existing data sets?		X
Section 5: ANIMALS		
Does your project involve animals?		X

Section 6: DEVELOPING COUNTRIES		
Does your project involve developing countries?		X
If your project involves low and/or lower-middle income countries, areany benefit-sharing actions planned?		X
Could the situation in the country put the individuals taking part in theproject at risk?		X
Section 7: ENVIRONMENTAL PROTECTION AND SAFETY		
Does your project involve the use of elements that may cause harm tothe environment, animals or plants?		X
Does your project deal with endangered fauna and/or flora /protectedareas?		X
Does your project involve the use of elements that may cause harm tohumans, including project staff?		X
Does your project involve other harmful materials or equipment, e.g.high-powered laser systems?		X
Section 8: DUAL USE		
Does your project have the potential for military applications?		X
Does your project have an exclusive civilian application focus?		X
Will your project use or produce goods or information that will require export licenses in accordance with legislation on dual use items?		X
Does your project affect current standards in military ethics - e.g., global ban on weapons of mass destruction, issues of proportionality, discrimina- tion of combatants and accountability in drone and autonomous roboticsdevelopments, incendiary or laser weapons?		X
Section 9: MISUSE		
Does your project have the potential for malevolent/criminal/terroristabuse?		X

Does your project involve information on/or the use of biological-, chemical-, nuclear/radiological-security sensitive materials and explosives, and means of their delivery?		X
Does your project involve the development of technologies or the creation of information that could have severe negative impacts on human rights standards (e.g. privacy, stigmatization, discrimination), if misapplied?		X
Does your project have the potential for terrorist or criminal abuse e.g. infrastructural vulnerability studies, cybersecurity related project?		X
SECTION 10: LEGAL ISSUES		
Will your project use or produce software for which there are copyrightlicensing implications?		X
Will your project use or produce goods or information for which thereare data protection, or other legal implications?		X
SECTION 11: OTHER ETHICS ISSUES		
Are there any other ethics issues that should be taken into consideration?		X